CYBER RANGE CAPSTONE

Cyber Defense Final Report

CORP-OSCORP

SOC incident assessment and host investigation package comparison

REPORT DATE	TIME STANDARD
10 October 2026	UTC unless explicitly stated otherwise

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Evidence-based laboratory assessment. Findings, evidence references and stated limitations retained from the supplied report.

01 SOC incident report

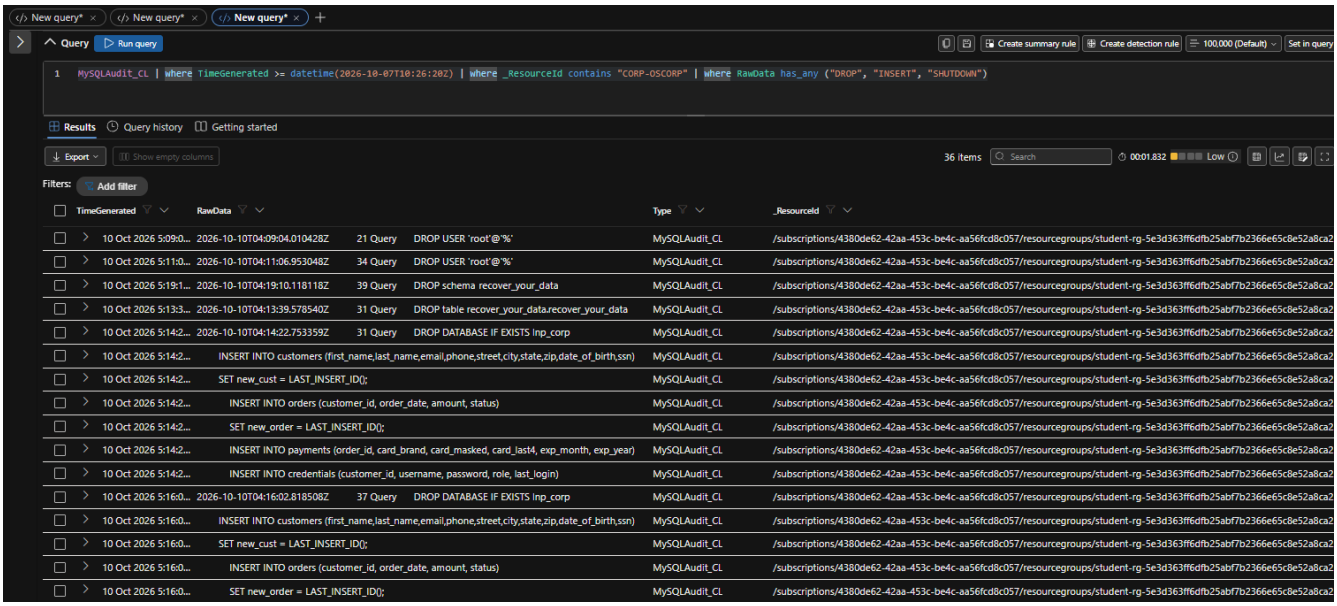
CORP-OSCORP | Windows and MySQL honeypot

10 October 2026 | UTC unless explicitly stated otherwise

1. Executive Summary

Following deliberate internet exposure on 7 October 2026, the CORP-OSCORP Windows/MySQL honeypot recorded database enumeration, submitted DROP DATABASE commands, and ransom-note insertion SQL. The database query log also shows attempts to disrupt binary logs and shut down MySQL; successful execution and data removal are not determined from available logs. Defender recorded a successful Administrator network logon from public IP 42.2.151.69 later that day, but its connection to the SQL activity is not established. This is a high-severity laboratory incident based on attempted destructive SQL and extortion; confirmed exfiltration, encryption and production impact are not determined from available logs.

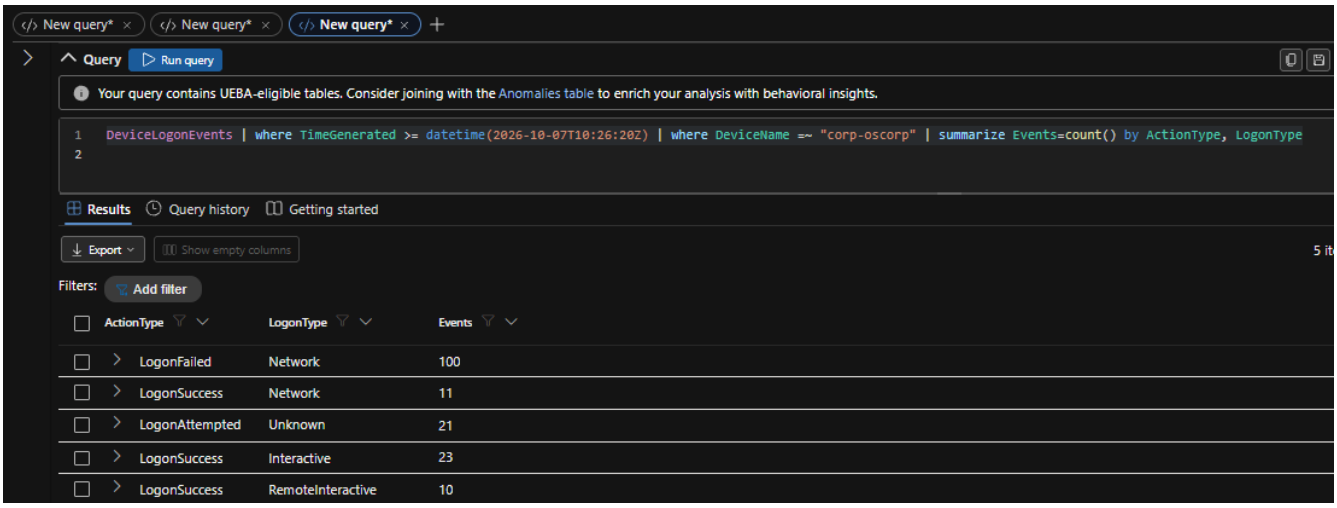
Original screenshot 01



2. Incident Details

Field	Evidence-based value
Detection / incident activity	Earliest destructive/extortion activity in provided SQL: 2026-10-07 11:53:33 UTC; original alert/detection time: Not determined from available logs
Reporter	Not determined from available logs
Type / severity	Destructive database commands and extortion-themed SQL / High (honeypot laboratory assessment)
Asset	CORP-OSCORP (Defender: corp-oscorp); Windows VM and MySQL server; LAW-Cyber-Range
Exposure start	2026-10-07 10:26:20.3547093 UTC (project checklist)
Isolation / closure	Not determined from available logs

Original screenshot 02



3. Impact Assessment

Confidentiality: SELECT queries against database tables were submitted; returned data volume or exfiltration is not determined from available logs. Integrity: DROP DATABASE commands targeted mika_corp, sakila and world; resulting deletions are not determined from available logs. Availability: SHUTDOWN, RESET MASTER and PURGE BINARY LOGS were submitted; actual outage or log loss is not determined from available logs. Scope is limited to observed activity on this honeypot; business or production impact is not determined from available logs.

Original screenshot 03

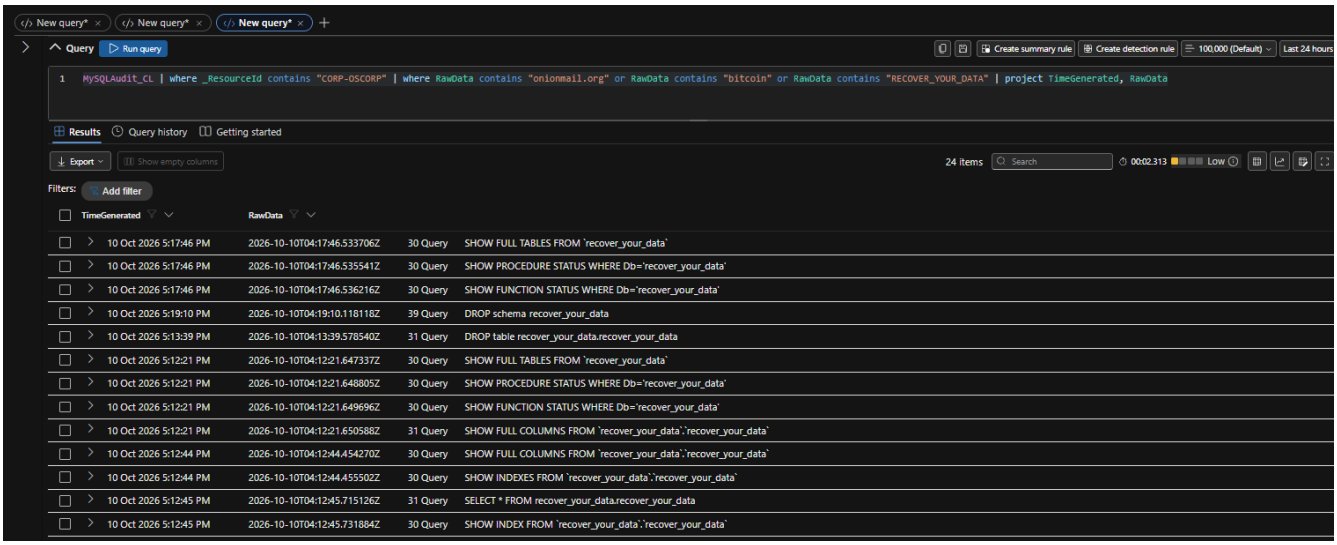


4. Indicators of Compromise

Indicator	Value	Source / confidence
Bitcoin wallet	bc1qt6kstjjdatqnxd3ksdcfv0jnhdjzndcnkx8zla	MySQL ransom INSERT / observed
Contact email	ak+2ijri@onionmail.org	MySQL ransom INSERT / observed
Ransom URL	https://cuq.in/844	MySQL ransom INSERT / observed
Ransom marker	RECOVER_YOUR_DATA	MySQL CREATE/INSERT / observed
DB auth source IPs	77.90.185.30; 64.89.163.94	MySQL auth CSV / observed
Windows logon IP	42.2.151.69	DeviceLogonEvents / observed successful network auth
Other user-provided BTC	bc1qk9kvwhzt60u3eqcjllqlj44h0tj7w7n72apz99	Not corroborated in supplied CSVs
Other user-provided email	ak+28t2@onionmail.org	Not corroborated in supplied CSVs
Other user-provided URL	hxxps://2no[.]co/2mysql	Not corroborated in supplied CSVs
Other user-provided DATAID	28T2	Not corroborated in supplied CSVs

These are observed indicators or leads, not proof that the corresponding IP addresses belong to the same actor.

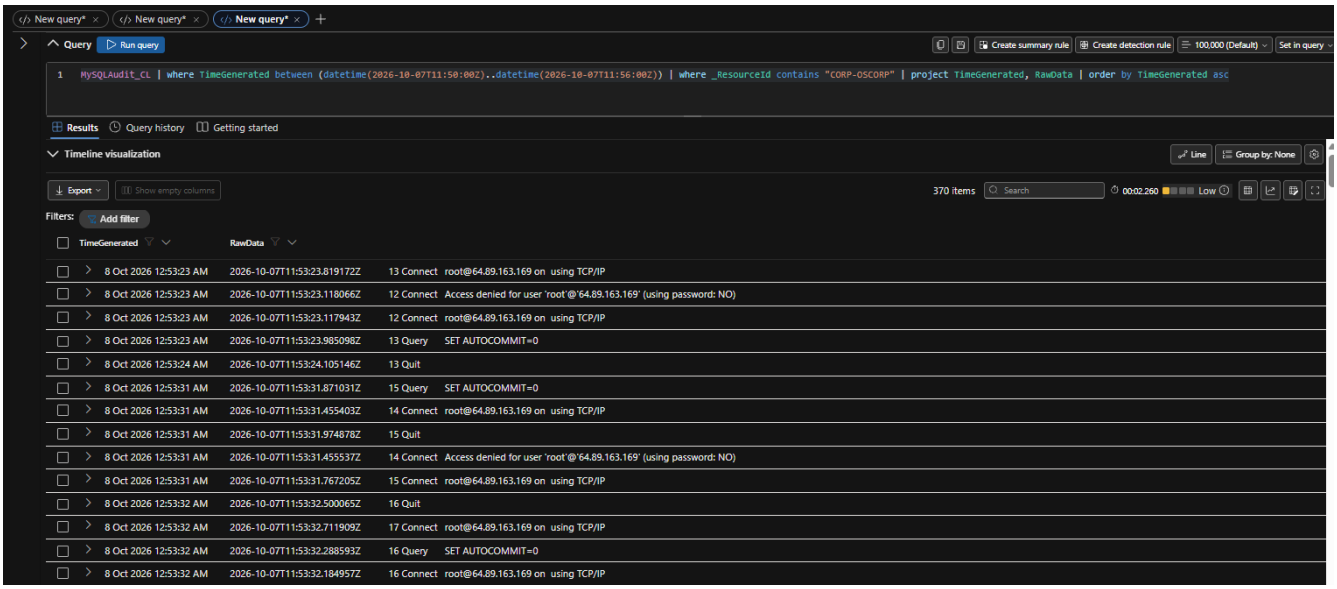
Original screenshot 04



5. Timeline

UTC	Event and source
07 Oct 10:26:20	Honeypot exposure, project checklist
07 Oct 11:53:33	Ransom-themed database INSERT attempts, MySQLAudit_CL_Query
07 Oct 11:53-11:54	Table/metadata SELECT and database discovery, MySQLAudit_CL_Query
07 Oct 11:54:12-13	DROP DATABASE mika_corp, sakila, world, MySQLAudit_CL_Query
07 Oct 11:54:13-14	CREATE/USE RECOVER_YOUR_DATA and ransom INSERT, MySQLAudit_CL_Query
07 Oct 11:54:15-17	RESET MASTER, PURGE BINARY LOGS, privilege statements, SHUTDOWN, MySQLAudit_CL_Query
07 Oct 19:31:28	Administrator LogonSuccess (Network), 42.2.151.69, DeviceLogonEvents
09 Oct 03:40-07:32	Repeated MySQL Connect and Access denied activity, MySQLAudit_CL_Auth
Detection time	Not determined from available logs

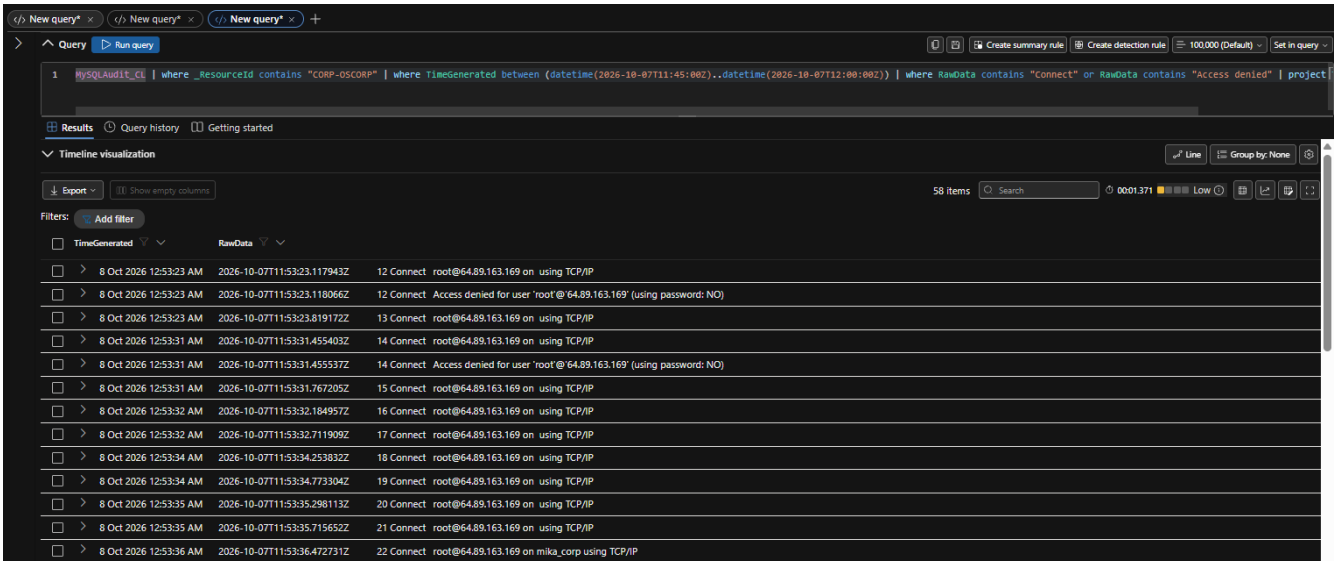
Original screenshot 05



6. Root Cause / Attack Vector

The honeypot was deliberately configured to accept internet-origin traffic. SQL execution is shown in the MySQL general-query log, but the exact database authentication session, originating IP, credential and entry path for the destructive commands are not determined from available logs. A separate successful Windows Administrator network logon is observed; a causal relationship to the MySQL attack is not determined from available logs.

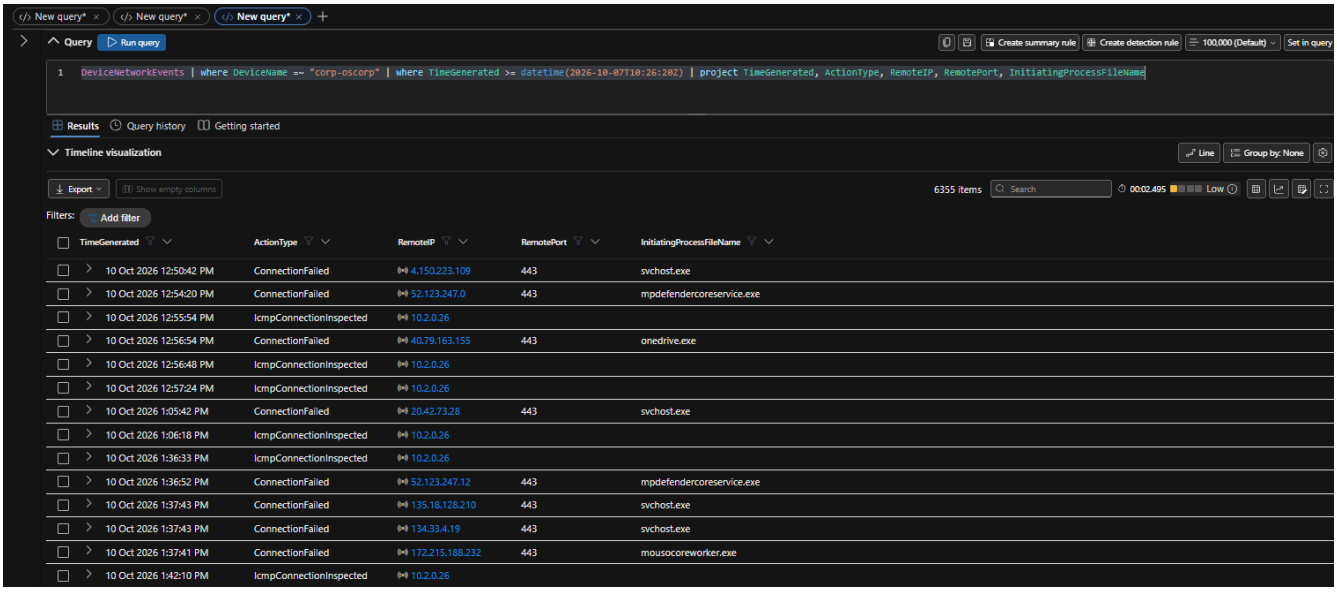
Original screenshot 06



7. Response Actions

Taken: monitoring and eight CSV evidence exports are reported by the analyst. Isolation, eradication, password resets, backup restore and recovery validation: Not determined from available logs. Recommended: isolate the VM in Defender; preserve pre/post isolation investigation packages and record UTC timestamps; restrict public RDP/MySQL exposure; rotate credentials; rebuild the intentionally weakened VM or restore MySQL from known-good backup; verify database contents, endpoint health and alert coverage before closure.

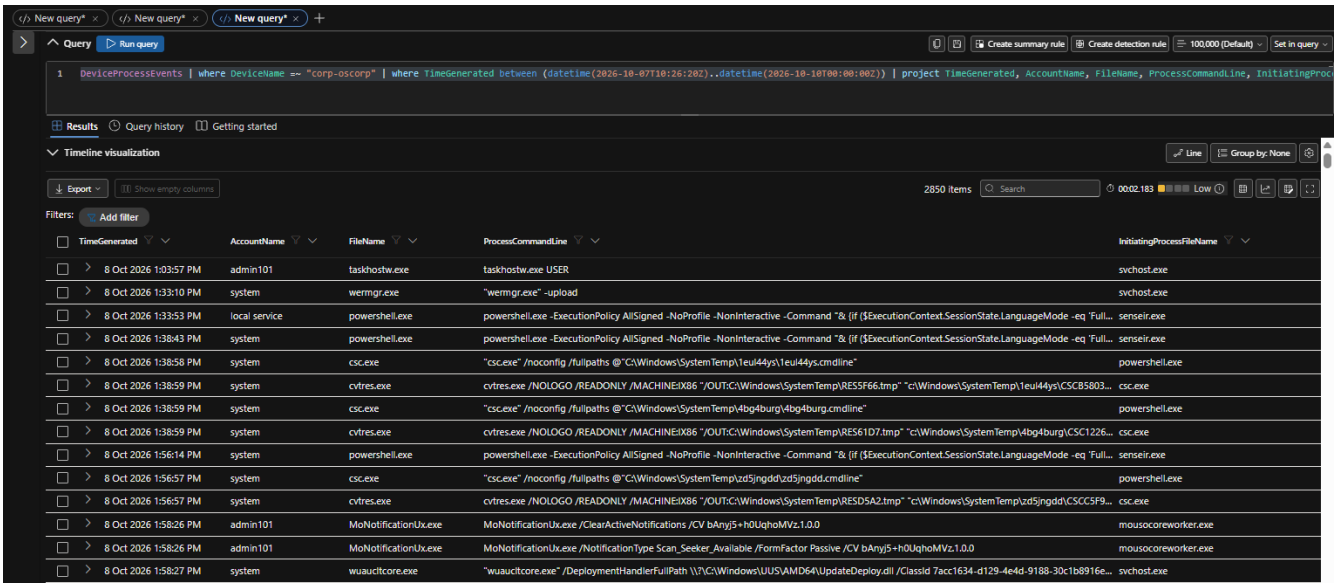
Original screenshot 07



8. Evidence

Reviewed: MySQLAudit_CL_Auth.csv (40 rows), MySQLAudit_CL_Query.csv (251 rows), DeviceLogonEvents.csv (155 rows), DeviceProcessEvents.csv, DeviceRegistryEvents.csv, DeviceNetworkEvents.csv, DeviceFileEvents.csv and NTANetAnalytics.csv (five exports with exactly 1,000 rows each). Also reviewed: the Cyber Range project checklist. Defender investigation packages before/after breach: not supplied. The five 1,000-row exports may be result-capped. MySQL authentication export covers 9 October and does not establish authentication for 7 October destructive SQL.

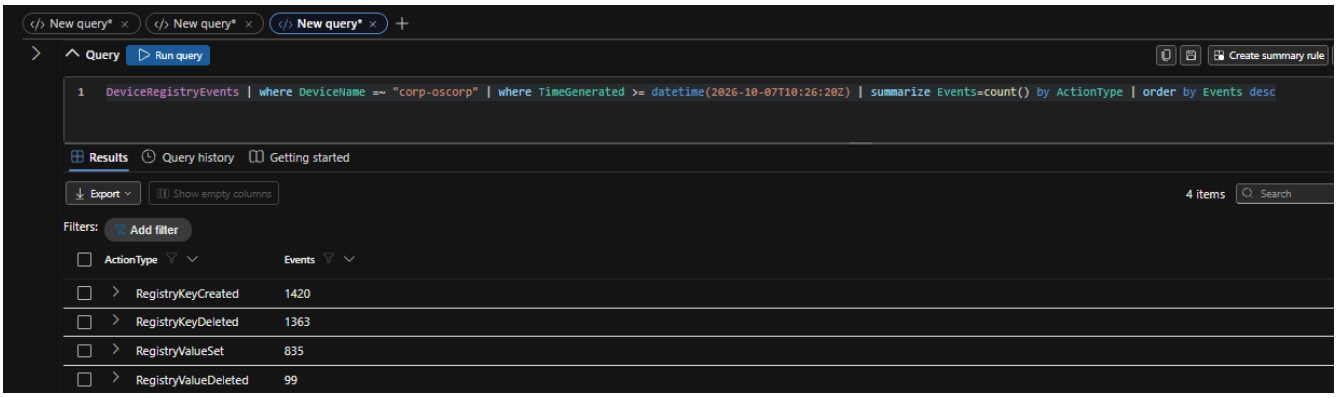
Original screenshot 08



9. Lessons Learned / Recommendations

- P1. Remove public RDP/3306 exposure; allow only approved management paths.
- P1. Replace weak/default credentials and disable unneeded accounts; rotate all exposed secrets.
- P1. Export complete, uncapped logs from exposure through documented isolation; retain original CSVs and hashes.
- P2. Correlate MySQL connection/thread IDs with source IPs and query sessions; alert on DROP DATABASE, ransom strings and privilege changes.
- P2. Validate tested MySQL backups and restoration; rebuild affected lab VM from a clean image.
- P2. Capture pre- and post-incident Defender investigation packages and document isolation/recovery times.

Original screenshot 09



02 Host DFIR report

CORP-OSCORP | 10.2.0.26 | MDE package comparison

10 October 2026 | UTC unless explicitly stated otherwise

Scope: Comparison of two MDE investigation packages, including nested Defender support archives.

1. Baseline determination

Role	Package	Internal collection timestamps
Baseline (B)	Pre-Breach_MDE_Investigation_Package (1).zip	7 October 2026, 10:32:21-10:33:10 UTC
Comparison (P)	Post-breach-package.zip	9 October 2026, 09:12:30-09:12:54 UTC

Collection order is established by each package's Forensics Collection Summary.csv, rather than ZIP modification dates. Both packages identify CORP-OSCORP. The earlier package is a comparison baseline, not a verified clean image.

Evidence notation: B = baseline; P = later package; RID = EVTX EventRecordID. Security.evtx means Security Event Log/Security.evtx. CAB artifacts are inside WdSupportLogs/MPSupportFiles.cab, under C_/ProgramData/Microsoft/Windows Defender/Support/. Text citations use line numbers; binary event logs use RIDs.

2. Summary verdict

Compromise indicators found: YES.

Confidence: High that suspicious access occurred; moderate that it was unauthorized. Two failed passwords were followed by a successful privileged-account network logon from a public IP and subsequent share enumeration. The packages cannot establish whether this connection was authorized.

3. Notable deltas

Category / change	Observed change and assessment	Evidence
Authentication / ADDED	42.2.151.69 fails twice, then authenticates as administrator; the session receives special privileges Assessment: Suspicious; strongest indicator	P Security.evtx, RIDs 42938-42942, events 4625/4672/4624
Discovery / ADDED	Same account/source invokes \PIPE\srvsvc, operation 15 Assessment: Suspicious share enumeration	P Security.evtx, RID 42943, event 5712
Authentication / ADDED	5,645 failed logons from 12 public IPs, plus 17 administrator lockouts; B has two failures without source IPs Assessment: Suspicious automated authentication attempts	P Security.evtx, example RIDs 40015, 49282 and 43887
Firewall / CHANGED	Domain, Public and Standard profiles: EnableFirewall=1 becomes 0 Assessment: Suspicious security weakening; actor and timing unresolved	B Autoruns/Autoruns.txt, lines 14609, 15134, 15675; P lines 14600, 15125, 15666
Application control / drivers / CHANGED / REMOVED	User-mode policy changes from Audit to Off; applockerfltr absent from later loaded-filter list Assessment: Suspicious control reduction, not proof of tampering	B/P System Information/SystemInformation.txt, line 52; B CAB FltmcInfo.txt, line 8, absent in P
Processes / integrity / ADDED	DefenderSessionHelper.exe now running; three invalid-image-hash events Assessment: Suspicious integrity anomaly; corruption or configuration issues remain possible	P Processes/Processes.csv, line 116; P Security RIDs 48950, 54640, 59198, event 5038
Autoruns / services / drivers / ADDED / CHANGED / REMOVED	DefenderUpdateSvc added; Defender driver path/version updated; per-user service instances replaced Assessment: Likely expected servicing/reboot activity; no clearly malicious persistence identified	P Services/Services.csv, line 34; B/P Autoruns/Autoruns.txt, lines 8869/8857; P CAB Microsoft-Windows-DefenderUpdateService%4Operational.evtx, RIDs 1-8, records update attempts
Scheduled tasks / CHANGED / REMOVED	OneDrive launcher version updated; SetupCleanupTask removed Assessment: Likely benign maintenance; no new suspicious task	B/P Scheduled Tasks/ScheduledTasks.csv, line 7; P Security RID 57416, event 4699
Network snapshot / DNS / UNCHANGED / CHANGED / REMOVED	Core listeners unchanged, including SMB 445, RDP 3389 and MySQL 3306/33060; RDP peer changes; DNS cache becomes empty Assessment: No demonstrated C2; session/cache changes are expected across reboots	B/P Network Connections/ActiveNetConnections.txt, lines 5-20; RDP peers at B line 41 and P line 44; Network Connections/DnsCache.txt
Users / groups / UNCHANGED	Administrators remain admin101 and administrator; memberships identical Assessment: No newly added privileged account between captures	Both Users and Groups/LocalGroups.txt, lines 19-20; files byte-identical

Category / change	Observed change and assessment	Evidence
File metadata / prefetch / browser / ADDED / CHANGED / UNCHANGED	47 additional prefetch files, none removed; temp-file churn; Edge cookie exporter already present Assessment: No confirmed malicious executable delta; cookie-exporter name alone is not credential-theft evidence	Both Prefetch Files/Prefetch/COOKIE_EXPORTER.EXE-DC47570F.pf: identical; references EdgeCore and an importer COOKIES.JSON
Event retention / ADDED / REMOVED	Security windows no longer overlap; System adds 344 records and Application adds 363, retaining baseline records unchanged Assessment: Retention gap, not proven clearing	Security RID ranges 1017-14510 in B and 37512-59470 in P; CAB system.evtx and application.evtx

Event 5038 can reflect malicious modification, software/configuration problems or corruption. The executable itself is needed to distinguish these possibilities. [1]

4. Prioritized indicators and ATT&CK mapping

1. Probable local-account misuse

At 7 October 2026, 19:31:28 UTC, 42.2.151.69 successfully authenticates using administrator, SID S-1-5-21-2406462958-2590928509-3634938128-1000, after two bad-password failures. Logon ID 0x221ee12 connects events 4624 and 4672. This account already existed and belonged to Administrators in B.

- Evidence: P Security.evtx, RIDs 42938-42942.
- ATT&CK: Initial access, T1078.003: Valid Accounts: Local Accounts. [2]
- Limit: Successful authentication is established; authorization is not. This is a type 3 network logon, not evidence of an interactive RDP session.

2. Share enumeration immediately afterward

At 19:31:30 UTC, the same account and source invoke SRVSVC operation 15, which Microsoft identifies as NetrShareEnum. [3]

- Evidence: P Security.evtx, RID 42943, event 5712, endpoint \PIPE\srvsvc.
- ATT&CK: Discovery, T1135: Network Share Discovery. [4]
- Limit: The request does not establish file access or exfiltration.

3. Large-scale authentication attempts

The later Security log contains:

Failure category	Count
Bad password	267
Account locked out	632
Requested logon type not granted	4,746
Total failed logons	5,645

These should not all be described as wrong passwords. There are also 17 account-lockout events for administrator.

- Evidence: P Security.evtx, event 4625; source-specific example RIDs in the IOC table below. Account lockout example: RID 43887, event 4740.
- ATT&CK: Credential access, probable T1110.001: Password Guessing. [5]

4. Firewall disabled across all profiles

Domain, Public and Standard firewall profile values change from enabled to disabled between captures.

- Evidence: B/P Autoruns/Autoruns.txt, line references in the delta table.
- ATT&CK: Defense evasion behavior, Disable or Modify System Firewall. Historical mapping: T1562.004; current Windows Host Firewall mapping: T1686.003. [6][7]
- Limit: The configuration change is established, but attribution to the successful remote session is not.

The App Control and image-integrity anomalies are secondary investigation leads, not independently proven attacker actions. A specific malicious ATT&CK technique is not established for those anomalies.

5. IOC derivation and cross-package hunt

Both packages were searched using decoded text, EVTX records, prefetch references and extracted CAB text/logs. Absence means absent from the supplied artifacts only.

Indicator	Baseline result	Later result / evidence
42.2.151.69	No match	Two failed passwords, successful administrator logon and share enumeration; Security RIDs 42938-42943
80.94.95.83	No match	3,794 failures; first Security RID 49282
23.94.69.14	No match	970 failures; first RID 40015

Indicator	Baseline result	Later result / evidence
180.94.20.203	No match	449 failures; first RID 48391
160.250.246.153	No match	186 failures; first RID 43788
172.216.11.33	No match	164 failures; first RID 40071
138.91.109.48	No match	36 failures; first RID 42878
80.66.83.43	No match	35 failures; first RID 53378
66.179.249.79	No match	6 failures; first RID 49144
45.156.128.176	No match	1 failure; RID 43316
66.181.39.227	No match	1 failure; RID 49362
47.251.30.24	No match	1 failure; RID 51168
administrator, SID ending 1000	Present and privileged	Used by the suspicious successful session; account name alone is not a malicious IOC
Session ID 0x221ee12	No match	Correlates the successful logon and special-privilege events
DefenderSessionHelper.exe	Already documented in CAB FileVersions.txt, line 16	Running, with new integrity failures; a validation target, not confirmed malware

No defensible malicious domain, payload hash or persistence-task IOC was established. These IPs are observed authentication sources, not demonstrated C2 destinations. The broader authentication activity is not proven to originate from a single actor.

6. Probable incident type

Probable local-account compromise through exposed network authentication, followed by SMB/RPC share discovery, amid broader password-guessing activity.

There is insufficient evidence to identify ransomware, data theft, malware deployment or lateral movement. The firewall change cannot be causally linked to the successful session. Whether the successful source IP was authorized remains unresolved.

7. Gaps and limitations

- Critical Security-log gap: Neither package contains RIDs 14511-37511, between approximately 7 October 10:32 and 18:50 UTC. Advancing record numbers fit rollover; no new clearing event was found. Historical System log clearing records were already present in the baseline.

- Execution visibility: Process-creation auditing is sparse and captured 4688 command lines are empty. There is no complete execution timeline, memory image or loaded-module inventory. Prefetch references do not substitute for a memory-based module assessment.
- Filesystem and payload verification: No full filesystem image, MFT/USN contents or suspicious executable samples were available. Executable integrity, signatures and malicious payload hashes could not be independently verified.
- Browser, credentials and hosts: No browser databases, credential stores or standalone hosts-file capture were available for comparison.
- Firewall traffic: Network Connections/FirewallExecutionLog.txt, line 3, reports pfirewall.log missing. Registry configuration was available, but traffic history was not.
- Attribution and authorization: Whether 42.2.151.69 was authorized, and who disabled the security controls: Not determined from available logs.
- Impact: Data access, exfiltration, destructive activity and downstream compromise: Not determined from available logs.

References

Package evidence is the basis for host-specific findings. The references below support event interpretation and ATT&CK classification only.

1. Microsoft: Event 5038, invalid image hash
2. MITRE ATT&CK: T1078.003, Local Accounts
3. Microsoft MS-SRVS: NetrShareEnum, Opnum 15
4. MITRE ATT&CK: T1135, Network Share Discovery
5. MITRE ATT&CK: T1110.001, Password Guessing
6. MITRE CTID: ATT&CK 17.1 T1562.004, Disable or Modify System Firewall
7. MITRE ATT&CK: T1686.003, Windows Host Firewall

03 KQL investigation appendix

Suggested investigation queries

Queries reproduced from the SOC incident report. Each query is associated with its original section.

1. Executive Summary

```
MySQLAudit_CL
| where TimeGenerated >= datetime(2026-10-07T10:26:20Z)
| where _ResourceId contains "CORP-OSCORP"
| where RawData has_any ("DROP", "INSERT", "SHUTDOWN")
```

2. Incident Details

```
DeviceLogonEvents
| where TimeGenerated >= datetime(2026-10-07T10:26:20Z)
| where DeviceName =~ "corp-oscorp"
| summarize Events=count() by ActionType, LogonType
```

3. Impact Assessment

```
MySQLAudit_CL
| where _ResourceId contains "CORP-OSCORP"
| where RawData contains "Query" and (RawData contains "DROP DATABASE" or RawData contains "SHUTDOWN"
or RawData contains "PURGE BINARY")
| project TimeGenerated, RawData
```

4. Indicators of Compromise

```
MySQLAudit_CL
| where _ResourceId contains "CORP-OSCORP"
| where RawData contains "onionmail.org" or RawData contains "bitcoin" or RawData contains
"RECOVER_YOUR_DATA"
| project TimeGenerated, RawData
```

5. Timeline

```
MySQLAudit_CL
| where TimeGenerated between (datetime(2026-10-07T11:50:00Z)..datetime(2026-10-07T11:56:00Z))
| where _ResourceId contains "CORP-OSCORP"
| project TimeGenerated, RawData
| order by TimeGenerated asc
```

6. Root Cause / Attack Vector

```
MySQLAudit_CL
| where _ResourceId contains "CORP-OSCORP"
| where TimeGenerated between (datetime(2026-10-07T11:45:00Z)..datetime(2026-10-07T12:00:00Z))
| where RawData contains "Connect" or RawData contains "Access denied"
| project TimeGenerated, RawData
```

7. Response Actions

```
DeviceNetworkEvents
| where DeviceName =~ "corp-oscorp"
| where TimeGenerated >= datetime(2026-10-07T10:26:20Z)
| project TimeGenerated, ActionType, RemoteIP, RemotePort, InitiatingProcessFileName
```

8. Evidence

```
DeviceProcessEvents
| where DeviceName =~ "corp-oscorp"
| where TimeGenerated between (datetime(2026-10-07T10:26:20Z)..datetime(2026-10-10T00:00:00Z))
| project TimeGenerated, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName
```

9. Lessons Learned / Recommendations

```
DeviceRegistryEvents
| where DeviceName =~ "corp-oscorp"
| where TimeGenerated >= datetime(2026-10-07T10:26:20Z)
| summarize Events=count() by ActionType
| order by Events desc
```